

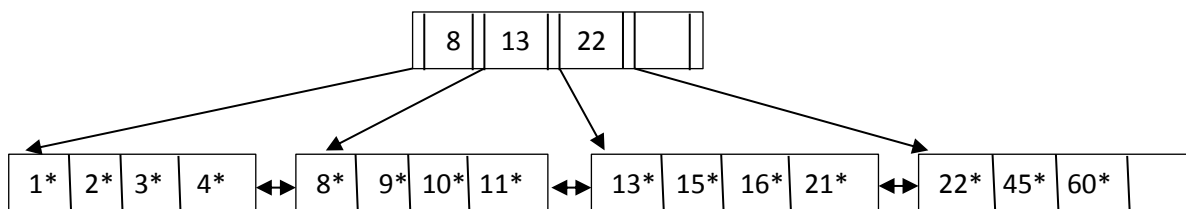
Homework # 2 – Solutions Set

1. Apply the bulk loading algorithm for the records having key values 10, 2, 1, 11, 16, 8, 3, 21, 22, 15, 9, 4, 45, 60, 13 to form a B+ tree of degree 2 (both for the index nodes and leaf nodes).

Answer: First of all you need to sort the data. Sorted key values are as follows:

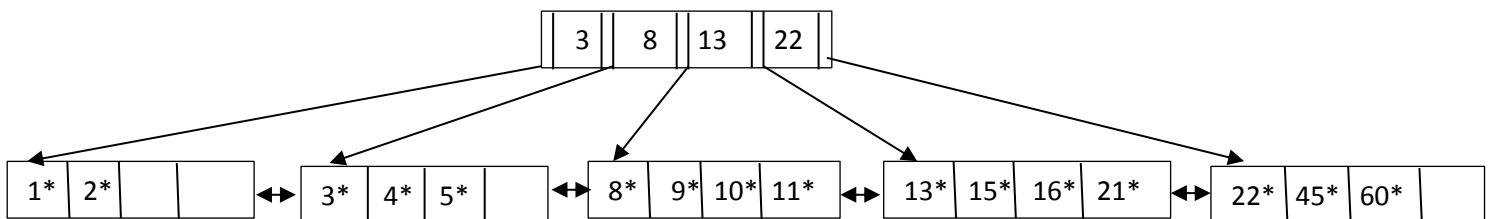
1, 2, 3, 4, 8, 9, 10, 11, 13, 15, 16, 21, 22, 45, 60

Since $d=2$, in each node (page) we have at least two and at most four keys. The below figure shows the B+ tree after applying the bulk loading algorithm.

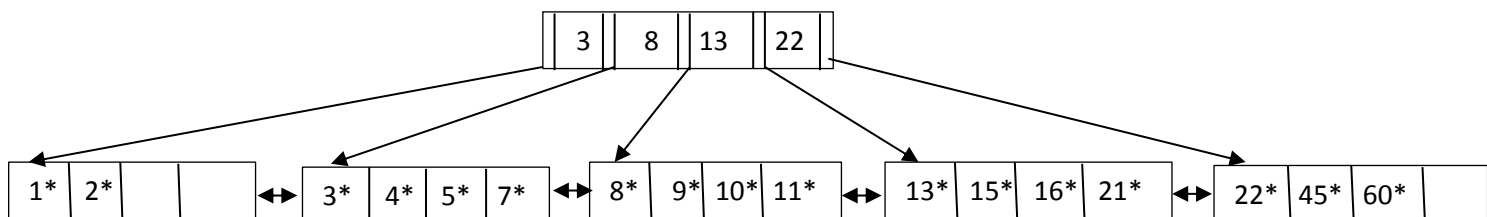


2. Insert 5 and 7 to the B+ tree that you have formed in the 1st question. And draw the B+ tree after each insertion.

Answer: The below figure shows the B+ tree after we insert 5:

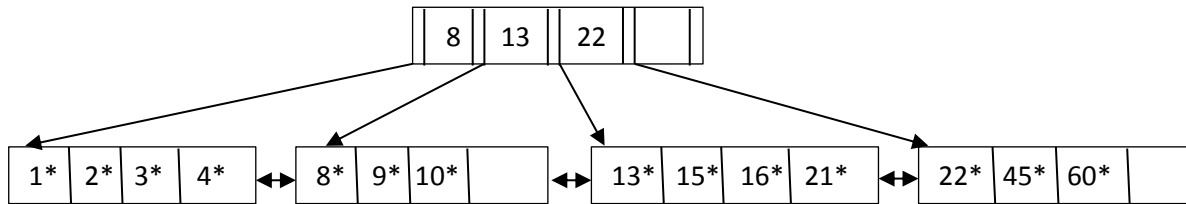


The below figure shows the B+ tree after we insert 7:

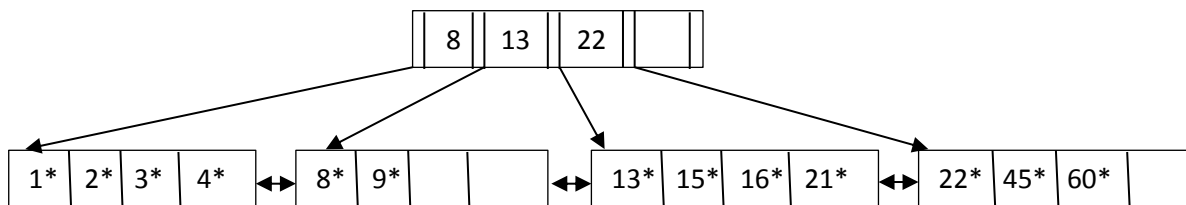


3. Delete 11, 10 and 9 from the B+ tree that you have formed in the 1st question. And draw the B+ tree after each deletion.

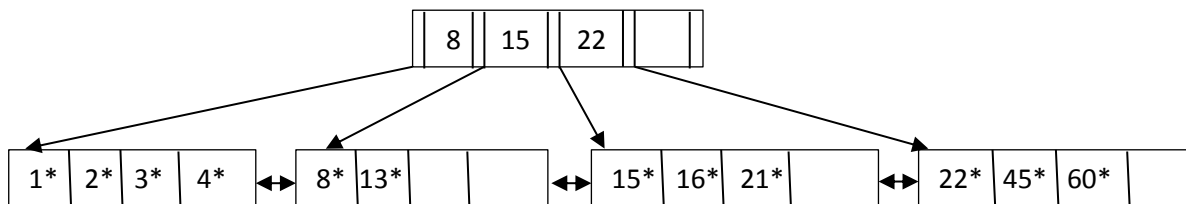
Answer: B+ tree after deleting 11.



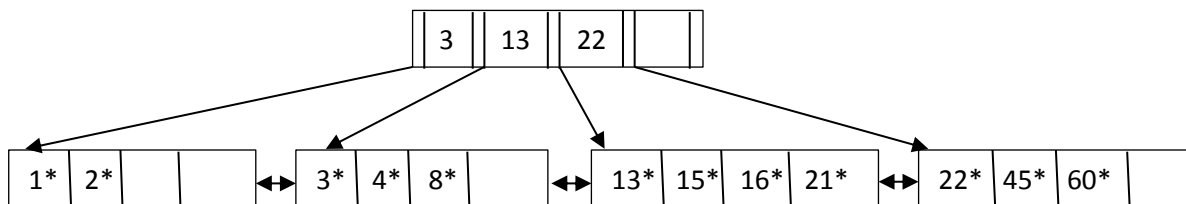
B+ tree after deleting 10.



B+ tree after deleting 9. In that case we need to rearrange keys between siblings. The below figure uses the right sibling.

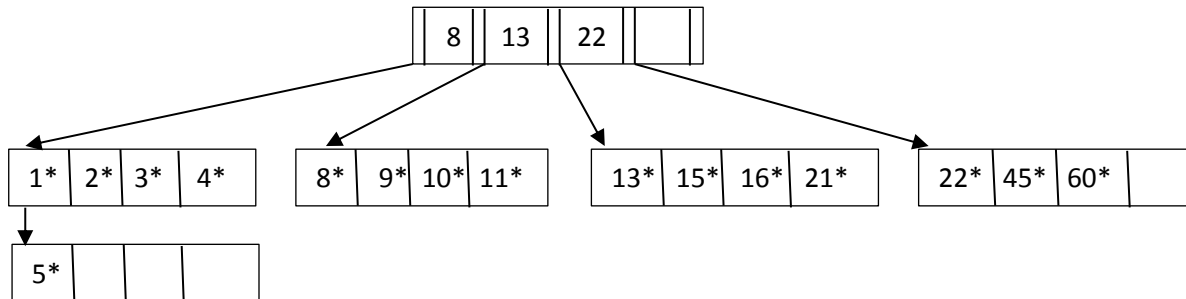


The below figure uses the left sibling.

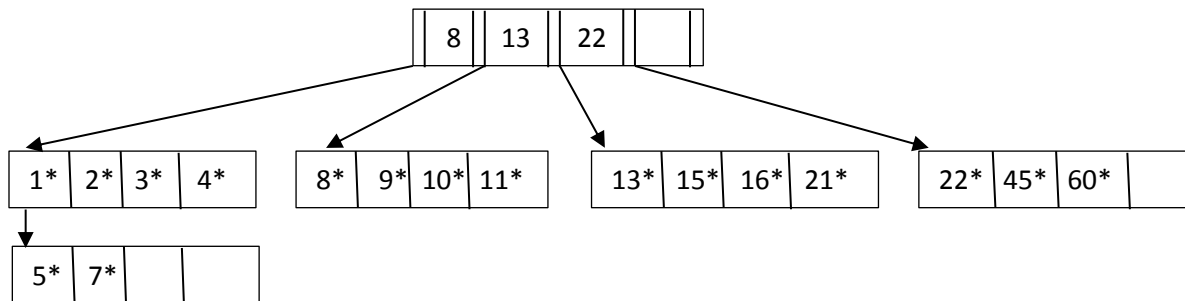


4. Assume that the tree index that you have drawn in the 1st question is an ISAM index. Then, insert 5 and 7 to the ISAM structure and draw the tree after each insertion.

Answer: The below figure shows the ISAM after we insert 5.

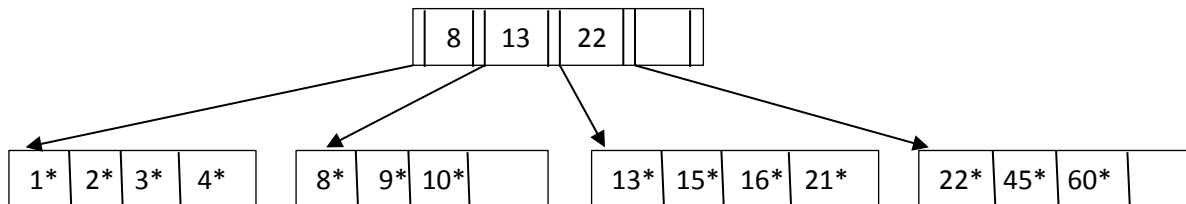


ISAM structure after we insert 7.

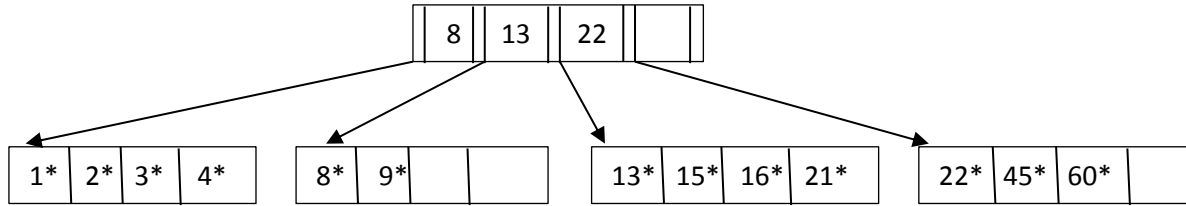


5. Assume that the tree index that you have drawn in the 1st question is an ISAM index. Then, delete 11, 10, and 9 from the ISAM structure and draw the tree after each deletion.

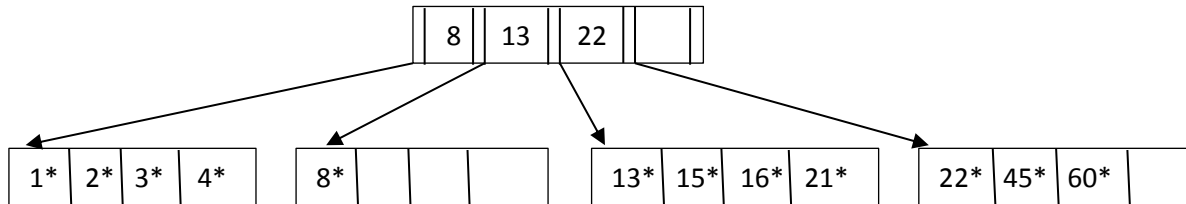
Answer: The below figure shows the ISAM after we delete 11.



The below figure shows the ISAM after we delete 10.



The below figure shows the ISAM after we delete 9.



Note that leaf nodes are not merged, nor rearranged, since ISAM is static.